Amazon Sales Dashboard

Data Analysis using Power BI

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Project Overview

Project Goal

Transform raw Amazon sales data into actionable business insights through interactive visualization

Dataset Source

Kaggle Amazon Sales Dataset containing comprehensive product information and customer metrics

Key Metrics

Product Name, Category, Price, Discount, Rating, and Reviews analyzed for strategic insights

This analysis demonstrates how data preparation and visualization tools can unlock valuable insights from e-commerce data, enabling informed business decisions.

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Data Preparation & Tools

Data Cleaning Process

- Handled missing values through systematic imputation and filtering
- Converted data types for consistency and optimal performance
- Created calculated columns including Discount % and Revenue metrics
- Standardized product categories and subcategories

Technologies Used

- **Power BI** for interactive dashboard creation
- **Power Query** for data transformation
- **DAX** for calculated measures and KPIs

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Key Performance Indicators

Total Sales

Sum of all revenue generated across products

Total Products

Unique products analyzed in the dataset

Average Rating

Mean customer satisfaction score across products

Average Discount

Typical discount percentage applied to products

Average Price

Mean price point across the product catalog

These KPIs provide the foundation for understanding overall business performance and identifying trends across product categories.

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Dashboard Overview

Summary Cards

Key metrics displayed prominently for quick performance assessment at a glance

Category Filter

Dynamic filtering by main and subcategories to drill down into specific product segments

Product Selector

Search and select specific products to analyze individual performance metrics

Date Range

Time-based analysis to track performance trends and seasonal patterns

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Product & Category Analysis

Top Products by Sales

High-performing products driving the most revenue, analyzed by both unit sales and total revenue contribution.

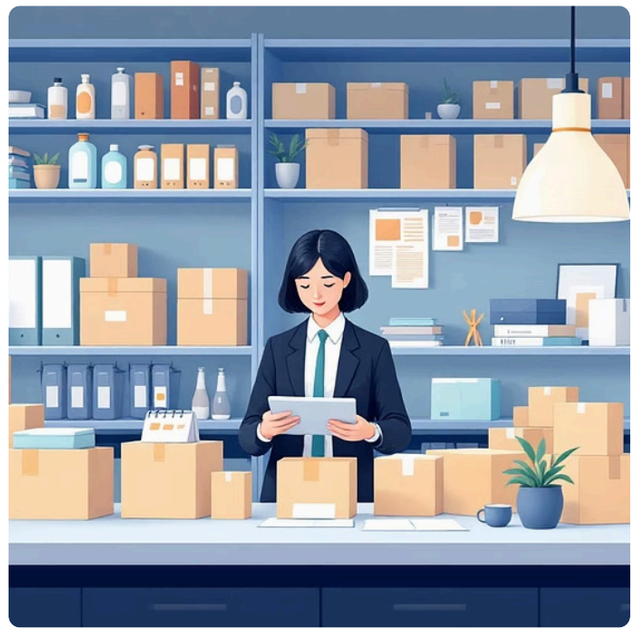
Sales Distribution

Understanding which products contribute most to overall revenue and identifying long-tail opportunities.

Category Performance

Main Categories

Sales distribution across broad product categories including Electronics, Computers, and Mobile Accessories



Subcategory Analysis

Detailed breakdown within each category to identify specific product types driving performance

Category Comparison

Cross-category insights revealing relative performance and strategic opportunities

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Key Business Insights

Electronics Dominance

Electronics category leads in sales volume, driven by high-demand products like smartphones and laptops

Discount Impact

Strategic discounting shows strong correlation with increased sales velocity across most categories

Rating Behavior

Average ratings cluster around 4.0-4.5 stars, with minimal correlation to price or discount levels

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Strategic Recommendations

01

Prioritize High-Performing Categories

Allocate marketing budget and inventory resources to Electronics and Computers categories where demand is strongest

02

Optimize Discount Strategies

Test tiered discount approaches to maximize revenue while maintaining profit margins

03

Address Low-Rated Products

Identify products with ratings below 3.5 stars for quality improvements or strategic discontinuation

04

Expand Product Portfolio

Consider adding complementary products in high-performing categories to increase average order value

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Conclusion

Summary of Findings

This analysis successfully transformed raw Amazon sales data into meaningful business insights through comprehensive data preparation and visualization techniques.

The dashboard enables stakeholders to quickly identify performance trends, compare category performance, and make data-driven decisions about inventory, pricing, and marketing strategies.

Value of Data-Driven Decisions

By leveraging Power BI's interactive capabilities, this dashboard provides ongoing value for monitoring business performance and testing strategic hypotheses.

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Thank You

Questions? Let's discuss how data analytics can drive your business forward.

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